XJTLU

XJTLU Entrepreneur College (Taicang)

Module Code and Title	ENT208TC Industry Readiness
School Title	XJTLU Entrepreneur College (Taicang)
Assignment Title	Validation Report
Submission Deadline	**Friday 15 May, 2026 before 23:59 (Beijing Time, GMT+8)**
Session and Group	[Session1-Group27]
Student IDs	2362452,2360861,2361326,2361210,2361098,2364028, 2362478,2361871
Final Word Count	1,100 words
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1st Marker — red pen							
Moderation — green pen	IM Initials	The original mark has been accepted by the moderator: Y / N					Y / N
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How to use this template

The sections below are a suggested structure — not a rigid requirement. Your report should tell the story of how you tested your product with real users and what you changed as a result. If your project used a different method or your findings are better organised another way, adapt the structure. What matters is that the marker can clearly see: who you tested with, what you found, and what you changed.

✓ Submission checklist — confirm before uploading
☑ 4–6 pages total (excluding appendix and references)
☑ Font: Calibri or Times New Roman, 12pt, 1.5 line spacing, justified
☑ File: Session[X]Group[Y]_ValidationReport.docx (.docx or .pdf)
☑ Cover page: group number + all 7 student IDs
☑ At least 5 user testing participants documented
☑ Direct quotes or observed behaviours included (not just summaries)
☑ At least one clear iteration shown: finding → change → evidence
☑ Before/after comparison for at least one change
☑ Evidence appendix: links to transcripts, notes, or recordings
☑ All evidence links are functional and accessible
☑ All references in APA 7th edition (not counted in page limit)

Section 1 — Who You Tested With & How

Your participants

Participant	Relevant background / why they were chosen	How you recruited them	Session format (in-person / online / etc.)
P1	International student; Regular gym-goer. Needed to test if the AI feedback terms (in English) were accurate for experienced users.	Directed invitation via WeChat.	Online (WeChat Chat / Video).
P2	International student; Fitness beginner. Vital for testing if the AI instructions were clear enough for someone without prior form knowledge.	Recruited via an international student WeChat group.	Online (WeChat Chat / Video).

P3	Chinese student; High-frequency user of fitness apps (e.g., Keep). Chosen to compare our scoring system against local market leaders.	Direct contact via WeChat.	Online (WeChat Chat / Video).
P4	Chinese student; Sedentary lifestyle / Office worker profile. To test if the app motivates "zero-base" users to start exercising.	Personal network recruitment.	Online (WeChat Chat / Video).
P5	Chinese student; Sports major. To verify the technical accuracy of the AI's "pose correction" feedback from a professional angle.	Directed invitation via WeChat.	Online (WeChat Chat / Video).
P6	Chinese student; Tech enthusiast. Focused on testing the UI responsiveness and the camera-tracking latency on different mobile devices.	Recruited via a tech-interest WeChat group.	Online (WeChat Chat / Video).

What you asked them to do

We asked each participant to complete the following tasks to evaluate the core AI functionality and user experience:

Task 1: Complete the initial setup and fitness profile to see if the app's difficulty levels were appropriate.

Task 2: Perform a set of 10 squats in front of the mobile camera to test the AI's real-time motion-tracking and pose correction.

Task 3: Read and interpret the "Correction Report" and "Final Score" provided by the AI after the exercise.

Task 4: Provide feedback on the clarity of the voice instructions and the visual feedback during the movement.

Sessions lasted approximately 15 – 20 minutes each.

How you captured what happened

The sessions were primarily conducted and recorded via WeChat. We captured the user experience through:

Chat Transcripts: Detailed text feedback from participants regarding their feelings and any difficulties encountered.

Screenshots: Images of the app's interface at specific moments when users felt confused or satisfied.

Video Clips: Screen recordings or videos sent by users showing the AI tracking their movements.

We paid close attention to where the AI failed to recognize movements and whether the users understood the scoring logic. All records are stored in the Evidence Appendix (see Appendix A1 – A6).

Section 2 — What You Found

Finding 1: Lack of Actionable Guidance and Specific Correction Data

Observation: Participants struggled to translate the AI's "wrong" notification into a physical correction. They felt the feedback was too general and lacked the quantitative data needed to understand the extent of their errors.

Direct Quotes: P2 said: "I can't really realize which specific joint or point went wrong and needs improvement."

P4 noted: "The AI feedback is quite abstract... It just tells me I'm wrong without data or analysis to show how much I was wrong or where exactly."

Finding 2: Limited Visualization of Continuous and Global Movement

Observation: The current system focuses heavily on individual joint points, which prevents users from seeing their overall body alignment. Furthermore, users found it difficult to analyze their form through static feedback rather than continuous motion capture.

Direct Quotes: P1 suggested: "I can't see an evaluation for my whole body. I think adding an overall assessment would be better."

P3 complained: "Static movements can't show me the continuous process of how I completed the exercise... and where is the force point?"

Finding 3: System Positioning: "Detection Tool" vs. "Intelligent Assistant"

Observation: Users, especially beginners, expect the app to act as a coach that guides their fitness journey over time. The absence of historical data and personalized workout suggestions makes the app feel like a one-off utility.

Direct Quotes: P5 stated: "I hope you can not only tell me what's wrong with my form, but also give me more exercise suggestions or guidance."

P6 expressed frustration: "No training records! I can't go back and check the previous questions anymore."

Section 3 — What You Changed

What finding triggered this change	What you changed (before → after)	Evidence that the change helped
Finding 1: Users felt AI feedback was "abstract" and lacked specific joint data or angle analysis (P2, P4).	Before: App only showed a "Wrong" or "Correct" label. After: Added real-time joint angle data (e.g., "Knee angle: 85°") and specific coaching tips.	In follow-up tests, P4 noted that seeing the specific angle helped them understand exactly how to correct their form.

Finding 2: Users couldn't see their whole body or continuous movement (P1, P3).	Before: Static joint detection points on a live feed. After: Implemented a "Whole Body Skeleton Overlay" and a 3D playback feature.	P1 confirmed that the overall assessment allowed them to check their full alignment more effectively.
Finding 3: The app lacked training records and progress tracking (P6).	Before: Results disappeared after the session ended. After: Added a "Training History" module where users can track scores over time.	P6 expressed high satisfaction, stating they felt more motivated to use the app daily to track their progress.
Finding 4: Users wanted to see the continuous process and where they should feel the force (P3).	Before: App only displayed static joint nodes and basic score labels. After: Added a "Heatmap Overlay" to show muscle engagement and a "Trajectory Line" to visualize the flow of the movement.	In follow-up discussions, P3 noted that the heatmap helped them understand which muscles to activate, making the AI feel more like a professional trainer.

Most significant change — explained

The most significant change was the transition from abstract notifications to data-driven coaching. Following feedback from P2 and P4, we realized that without quantitative data (like joint angles), users were left confused. We redesigned the interface to include a Real-Time Data Panel. This shows the specific degree of flexion in the joints and uses color-coded arrows (Red for incorrect, Green for correct) to guide the user.

What you chose not to change — and why

We decided not to implement Real-Time Human Coaching. While P5 suggested more personalized guidance, our goal is to maintain a fully automated AI tool that is available 24/7. Instead, we improved the AI Recommendation Engine to provide automated workout plans, addressing the need for guidance while keeping the product scalable.

Section 4 — Limitations

• **Sample size:** How many people did you test with? What kinds of users could you not reach?

• **Testing conditions:** Were sessions realistic? What was different from real-world use?

• **What you could not test:** Were there features or scenarios you were not able to test?

Sample size: We tested our product with a small sample of 6 participants, primarily consisting of university students. This limited sample size means we could not reach a broader demographic, such as elderly users or professional athletes, whose specific fitness needs and physical limitations might differ significantly from our current test group.

Testing conditions: The testing sessions were conducted remotely via WeChat, which is different from a real-world gym environment. We could not control environmental factors such as lighting, background clutter, or the stability of the participants' mobile devices. These variables can affect the AI's motion-tracking accuracy and may not fully reflect how the app performs under optimal or standardized conditions.

What you could not test: Due to the one-time nature of our validation sessions, we were unable to test long-term user engagement or the effectiveness of the "Training History" and progress-tracking features. Furthermore, we could not test the app's performance on a wide variety of mobile hardware to assess potential latency issues across different processor speeds and camera qualities.

Section 5 — Evidence Appendix

Reference	What it contains	Link
A1	WeChat chat log / Interview transcript — P1 (International)	https://drive.google.com/file/d/1sUFYXD7I80eRN_QzPKSITuLAlv9NflmE/view?usp=drive_link
A2	WeChat chat log / Interview transcript — P2 (International)	https://drive.google.com/file/d/1MMj_9p03BmCaArKc8syNe_i7FiKNpK2E/view?usp=drive_link
A3	WeChat chat log / Interview transcript — P3 (Chinese - Tiger)	https://drive.google.com/file/d/1KIwNe5AcU8c35AJPA-WkGcJyD669eoKh/view?usp=drive_link
A4	WeChat chat log / Interview transcript — P4 (Chinese)	https://drive.google.com/file/d/15EsPwe3MHydbZRe9rRr0_76ZPOjWZ6rn/view?usp=drive_link
A5	WeChat chat log / Interview transcript — P5 (Chinese - Summer)	https://drive.google.com/file/d/1qECVgT08rMXibsIrQxqFsqnfltqiVSs-/view?usp=drive_link

A6	WeChat chat log / Interview transcript — P6 (Chinese - Ricardo)	https://drive.google.com/file/d/1YBA6BtEXwLDxdDGcV9UffdNLhgFYVYjd/view?usp=drive_link
B1	N/A (Only chat-based feedback was collected for this stage).	N/A
B2	Observation notes and user feedback summary	N/A
C1	N/A (Only chat-based feedback was collected for this stage).	N/A
C2	Screenshots of the AI fitness app prototype during testing	N/A

References

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